



Academy of Liberal Arts



**The influence of spell checkers on the ability of AUC
students to produce corrections for spelling errors**

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Abstract

This research investigates how a spell checker influences the possibility of The American University in Cairo's students becoming independent in the correction of spelling errors. This research focused on those students who have already taken the first English critical writing course to see if their reliance on spell checkers prevented them from learning and correcting spelling mistakes independently. A quasi-experimental research design was used. Ten students participated, with five students in the control group and five students in the experimental group. The results of the pre-test, treatment, and post-test phases revealed that while spell checkers improved the immediate accuracy of spelling, students after using such technologies were able to correct homophones and phonetics misspells, yet committed repetitions of common misspelled errors. This showed a great reliance on the tool and poor independent spelling skills. Results also indicated the cross-linguistic influence on spelling correctly in English. The effect of this research paper supplies knowledge on the necessity to focus more on developing student's ability to spell correctly and follows the impact of technology and artificial intelligence on their linguistics competencies.

Keywords: Spell Checkers; Spelling Errors; Linguistic Skills; Academic Writing; Grammarly; AUC; Quasi-experimental Design.

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Declaration

I declare that the work in this research paper was carried out in accordance with the regulations of the American University in Cairo. This thesis has been composed solely by myself except where stated otherwise by reference or acknowledgment. It has not been previously submitted, in whole or in part, to this or any other institution for a degree, diploma or other qualifications.

Signed: Wassim Bousbia

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Chapter 1: Introduction

1.0 Introduction :

Spell checkers have therefore become common tools in academic writing. They correct mistakes and enable more accurate writing. However, there is a concern about how much students are reliant on these tools and whether such reliance affects their ability to spell correctly without the assistance of technology. This chapter discusses how spell checkers can impact students' spelling skills at the American University in Cairo (AUC). It covers the context of the study, definition of terms, problem statement, the aim of the study, research approach, and thesis overview.

1.1 Contextualizing the study:

The use of spell checkers within academic writing has been both praised and criticized in various studies. Some researchers, like Fouad et al. (2022) and Lin et al. (2017), found that long-term reliance on the spell checker will affect the students' independent spelling ability and therefore lead to the repetition of the same mistakes over and over again. These studies all argue that spell checkers minimize misspellings but are not effective in the long run in teaching correct spelling to learners. On the other hand, some literature, such as Huang et al. (2020) and Pakpahan (2022), put across the idea that spell checkers help students learn how to write correctly. These studies emphasize that spell checkers can provide immediate feedback and learning opportunities to enhance spelling ability over time. Despite these different opinions, there is a general agreement that emphasizes that spelling is crucial for linguistic competence, reading accuracy, comprehension, and fluency, says Reed (2012). Proper spelling mirrors an even deeper understanding of language mechanics, not to mention phonology and morphology.

1.2 Definition of Terms:

Spellchecker: A computer program that checks every word in any kind of electronic document or text and rectifies misspelled spelling with the right sequence of letters (Kuc, 2015).

Technology: Techniques, tools, machinery, systems, and scientific knowledge acquired for applying technical methods in a practical way to solve problems and their mutual influence on the environment with life and society (Cox, 2006).

Word Processor: An application, device, or machine used for creating, editing, retrieving, and printing text or documents (Knapp, 1986).

Writing: The process of applying characters, like letters, numbers, etc... in order to communicate thoughts and ideas in a readable form (Daniels and Bright, 1996).

Misspelling: Writing words with incorrect letters or an incorrect arrangement of letters; in other words, spelling a word incorrectly (Toole & Heift, 2002).

1.3 Problem statement, Gaps, and Research rationale:

It is generally accepted that spell checkers have taken academic writing to a higher level since they increase accuracy and expertise while saving years of proofreading time. These tools correct mistakes with suggestions so that students can generate more convenient documents. Spelling skills are crucial since they represent linguistic competencies and assist in reading accurately by integrating indispensable linguistic info, which includes phonology and morphology. - (Reed, 2012). Even more, spelling represents knowledge for the correct decoding of words, essential for reading comprehension and fluency. While students nowadays are so reliant on spell checkers, the question at stake is: How good spellers are students, and how much do they gain from the suggestions provided by spell checkers? There is a growing concern that over-reliance on spell checkers is going to prevent learners from learning and remembering correct spelling conventions, therefore, making similar mistakes repeatedly and developing only superficial knowledge of the mechanics of language. In this case, one may wonder how far technology should be

utilized and how much spelling skills should be taught and developed to function properly across academic fields. Within its core curriculum, the American University in Cairo (AUC) offers a wide variety of English courses designed to help students become proficient writers of academic English on their own. The extensive usage of spell checkers like Grammarly, however, begs questions about their true ability to write and spell correctly without the aid of technology. By examining how much AUC students rely on spell checkers and how this dependence might impact their basic writing and spelling skills, this study aims to close the research gap in the literature. By looking at this matter, the study looks forward to assessing how well the AUC English curriculum works and offer suggestions for improving students' ability to write independently. The results may play a significant role in guiding the creation of curricula, instructional strategies, teaching practices, and the integration of technology in academic writing instruction.

1.4 Thesis aim and objectives

The research aims to identify the impact of spell checkers on the ability of American University in Cairo (AUC) students -who took the Critical Writing course RHET1010- to correct spelling errors independently. It investigated whether depending on writing with spell checkers helps students learn to spell or not. This paper assumed that depending on spell checkers does not help AUC students learn to spell, therefore, they will commit the same errors without using the spell checker. While AUC aims to develop its students' English proficiency and offers many courses to build their language skills, there has been no research ascertaining whether their students can spell accurately well without the use of spell-checking. This is relevant, given the fact that the use of spell checkers may not give the true picture of their spelling proficiency. To investigate this further, this paper inquired and researched how well the students of AUC could spell if they were to do it on their initiative. The research will significantly fill this gap of the unknown area regarding their real spelling skills, and it will shape if AUC has to offer more support to students to get their spelling off technological dependence. Therefore, this Research paper aim to investigate the following research question: How does using a spell checker influence AUC students' ability to correct spelling mistakes?

1.5 Research approach:

The participants are AUC students who have completed the RHET 1010 course and are English as Foreign Language (EFL) learners; their education level will be up to the sophomore year. A purposive sample approach has been used to obtain 5 participants for the control group and 5 others for the experimental group. The participants took a quasi-experimental spelling dictation test; the researcher dictated 4 phrases, including some of the words that are commonly spelled incorrectly. The dictated words and phrases were repeated as required to ensure proper comprehension. The control group received two dictations without the spell checker, while the experimental group received three dictations: the first and third without the spell checker, and the second using Grammarly. The design allowed the comparison of measurement results between the pre-test, treatment, and post-test effects through the use of methods in linguistic and content analysis of the spelling accuracy of a spell checker.

1.6 Thesis Overview:

Chapter 2: Literature Review

This chapter reviews the existing literature on the use of spell checkers and their impact on students' spelling abilities. It examines various studies that have explored the positive and negative effects of spell checkers on writing accuracy and long-term spelling skills. The literature review provides a comprehensive background, setting the context for the current research.

Chapter 3: Methodology

The methodology chapter details the research design, sampling method, and procedures used to conduct the study. It describes the quasi-experimental spelling

dictation test administered to AUC students, including the criteria for selecting participants and the phases of the experiment. Ethical considerations and the data collection methods are also discussed.

Chapter 4: Results

This chapter presents the findings from the quasi-experimental spelling dictation test. It includes statistical data, graphs, and charts to illustrate the performance of the experimental and control groups. The results show a significant improvement in spelling accuracy for the experimental group during the treatment phase, with a slight decrease in the post-test phase. The control group's performance remained relatively stable.

Chapter 5: Discussion

The discussion chapter interprets the results, comparing them with findings from previous studies. It highlights the immediate benefits of using spell checkers and the limitations in terms of long-term spelling retention. The chapter also discusses the common challenges faced by students, such as phonetic errors and the influence of linguistic backgrounds on spelling performance, and highlights the cross-linguistic effects on spelling abilities.

Chapter 6: Conclusion

The conclusion summarizes the key findings and their implications for educational practices. It emphasizes the need for a balanced approach that combines the use of spell checkers with efforts to develop independent spelling skills. The chapter also suggests directions for future research, including longer-term studies and the exploration of different spelling interventions.

Chapter 2: Literature Review

2.1 Automatic spell checkers:

Automatic spell checkers have become crucial tools for students to ensure writing properly for the personal, public, or academic world. They are integrated with a wide range such as word processors, devices' keyboards, and web browsers, which emphasizes the role of spelling checkers in the student's life. While these tools offer convenience and time-saving, several pieces of literature investigated their impact on students' ability to improve spelling independently and correctly. Spelling requires putting effort and concentration into the word written to be correctly spelled and memorized; therefore, Fouad et al. (2022), and Lin et al. (2017) focused on the students and their ability to spell independently, and found that students who rely heavily on spell checkers face challenges to spell correctly individually. Additionally, other pieces of literature affirmed that using Spell checkers does not develop the student's abilities to spell correctly; thus, they occur the same spelling error again (O'Rourke et al., 2020; Rimbar, H., 2017). At the American University in Cairo, where English is the primary language, students often rely on spell checkers to write their papers. This reliance shadowed the true level of students in writing Academic English properly. Therefore, this paper investigates how using spell checkers influences AUC students' ability to correct spelling mistakes and assumes that using spell checkers does not help AUC students to correct spelling mistakes. The studies are inspected to eight studies to verify the hypothesis.

2.2 Historical Integration of spell checkers in writing:

The integration of spell checkers into writing practices has significantly evolved, greatly developing the quality of academic writing. The spelling checker is a function that aims to help students type correctly, by first identifying the error and then correcting grammar and typographical errors to come up with good writing (Lin et Al., 2017). However, it took time to reach the final function of the spell checkers as we know it today. According to Peterson (1980), the first spell-checker called SPELL was written as a program for English texts in 1961, yet the spell-checkers were not

available on computers until 1980, and with the development and improvement of this technology, it became widespread and used in my fields including academic writing. The integration of spell checkers in academic writing became obvious when it came to comparing the writing before and after integration in 1985. Hussein, N. (2024) conducted a quantitative research study to illustrate the integration of spell checkers and their impact on writing quality. By selecting and comparing academic texts from the University of North Texas, five of them were before the invention of the spell checker in mainstream computers and five of them after, noting that the texts were written for the same activity/purpose. The results show that the misspelling rate in sample texts had reduced from 43% in the first texts to 7% in the text written with a spell checker, indicating the significant effect of integration of spell checkers in academic writing.

2.3 Spell checkers and learning process:

2.3.1 Positive impact:

The students' usage of spell checkers for academic writing has shown positive aspects in developing their writing and spelling skills. Huang et al., (2020) and Pakpahan (2022) found that using Grammarly (a modern spell checker) has improved English Foreign Language (EFL) students' ability to write academically. As far as data collection methods are concerned, both research studies' methods relied on experiments and writing tests on +20 university students to measure the impact of using Grammarly on their writing. Pakpahan (2022) provided the most credible results on the impact of using spell checker on students' writing because the research conducted compared the control and experiment groups before and after using Grammarly, unlike Huang et al., (2020) who lacked a control group that represents the casual writing of students without Grammarly in the second experiment, which limited the study.

2.3.2 Negative impact:

On the other hand, several literatures affirm that using spelling checkers does not necessarily develop students' writing and spelling skills. Claiming that spelling is a long process that requires effort and concentration to learn; the spelling checker

gives quick suggestions that do not allow students to memorize it or learn from it, so they will repeat the same error several times (Lin et al., 2017; O'Rourke et al., 2020). These two studies tested the performance of university students before and after using spell checkers for academic writing following controlled experiment methods. While the results of O'Rourke et al., (2020) were insightful and showed that spell checkers do not develop students' abilities, the 1-year age gap between controlled and experiment groups in a sample of 36 university students affected the result because during that year gap, the experiment group have taken an English course unlike the control group, which may affect the experiment's results, unlike the control group the research conducted by Lin et al., (2017) where although the sample was bigger (88 university students), they limited educational and demographic differences between the control and experiment group to ensure more accurate results.

2.4 Similar findings:

One research paper that matches the objective of this study was conducted by Rimbar (2017). The author aimed to investigate if the students improved their spelling abilities by the correction suggested by the spell-checker tool. The method followed was experimental, a sample of 30 English learners was divided into a control group of 16 and an experimental group of 14. Both groups were asked to write the same paragraph (a) without a spell checker, then the experiment group was allowed to review their writing using a spell checker, and for the third writing, both groups were asked to write the same paragraph (a) without a spell checker. The analysis and discussion of results conclude that spell checkers did not help to improve students' writing because the mean of the experiment group after using the spell checker was closer to the results before using the spell checker. This research study has conducted a good experiment that answers the research question and proves the hypothesis, the external factors were controlled very well to eliminate any inaccurate results, however, the experiment duration is long, and it may affect the results of the experiment due to being exposed to other corrective feedback.

Another study that has been conducted by Margo et al. (2021) on the contributions of Short term memory (STM) and long term memory (LTM) to spelling abilities in French-speaking students provides insightful information. Strong data on the impact of STM on spelling accuracy is provided. However, generalizability to other languages is limited by concentrating on the 114 who understand French. While STM/LTL and spelling are correlated, the study did not thoroughly investigate the underlying factors. Additionally, increasing the diversity of the sample and looking at additional aspects like phonological awareness might provide a more thorough knowledge of spelling development.

2.5 Gap in the literature:

The American University in Cairo provides a wide range of English courses in the core curriculum to develop students' abilities to write proper academic English individually. However, the wide range of spell checkers in the university such as Grammarly raises the question of the ability of AUC students to spell and write properly without depending on the spell checkers. There is insufficient research and data that investigates this question and reflects the student's level.

Conclusion

To conclude, the primary aim of the eight resources is to investigate the impact of spell checkers on university student's ability to spell correctly. Although much literature has shown the positive impact of using spell checkers on university students, by saving time and ensuring good writing, depending on such tools for academic writing cannot improve writing skills and does not reflect the real level of the students who use spell checkers. And since AUC is an English institution that cares about the level of their students and their academic English writing abilities, it is crucial to know the real level of students without using spell checkers.

Chapter 3: Methodology

3.1 Ethical standards:

It is crucial to consider the ethical dimensions of a research study, and the methodology followed to reach the research objectives. Since this research experiment had a human sample, participants had the right to debriefing, which ensured that participants understood how participating in this study is helpful, and will be used for scientific purposes (Aguinis et al., 2004), as they were informed that these tests are conducted in a research paper for rhetoric and composition 1020 course to see to what extent spell checkers help AUC students correct spelling errors. As the rights of freedom were also respected, They had the right to cancel or stop the test during any phase of the 3 phases of the test, and they could cancel their previous answers to not be used in the results. The experiment test was done at AUC, which represents a safe academic environment for such an academic experiment, the tests had taken place in classes or common areas of dorms, with totally appropriate health and safety conditions.

3.2 Sample:

The sampling method followed was the purposive method, which is a group of different non-probability sampling techniques. Also known as judgmental sampling, or subjective sampling, relies on the researcher's judgment in selecting the units of study. Typically, the samples examined are very small, especially when compared to probability sampling techniques (Rai & Thapa, 2015). Two groups of AUC students were selected, the first group represented the control group (n=5), and the second group represented the experimental group (n=5). The selected participants must have passed the critical writing course before taking the test, noting that this course aims to develop students' writing skills. On top of that, the student's education level must not exceed the sophomore year. Furthermore, all the participants must be English as foreign language (EFL) students. These criteria aimed to limit the differences in spelling skills and abilities among the participants.

3.3 Method:

The participants received a quasi-experimental spelling dictation test in English, where they spelled Four dictated phrases face-to-face with a supervisor. The dictated phrases were well formed and 17 words among them which will be analyzed, are considered among the most frequently spelled words according to Oxford University Press (2012). To make sure that all the errors are spelling errors and not committed due to mishearing or lexical concerns, the words were put in phrases so that they were formed in meaningful frames in terms of context, meaning that it eliminated lexical errors such as the homophones. To exclude all external factors that may affect the accuracy of results, the participants are allowed to ask the supervisor to repeat and clarify the word dictated as much as they need, noting that the dictation time is unlimited. Finally, all the tests were computer-based with and without using spellchecker.

3.4 Procedure:

Noting that the phrases were the same in all phases, The control group were given two dictations without a spell checker, and they had the right to revise their writings after each writing; due to the short period of the experiment, they wrote the second time immediately after revising the first writing. The experiment group had three dictations, the first one was without a spell checker, and they had time to review it carefully similarly to the control group. The second dictation used a Grammarly free-version extension, which suggested misspelling corrections. After that, they wrote again for the third time without a spellchecker; This phase is important because it indicated to what extent the students benefited from the spellchecker corrections. The answers were compared, analyzed and summarized.

Chapter 4: Results

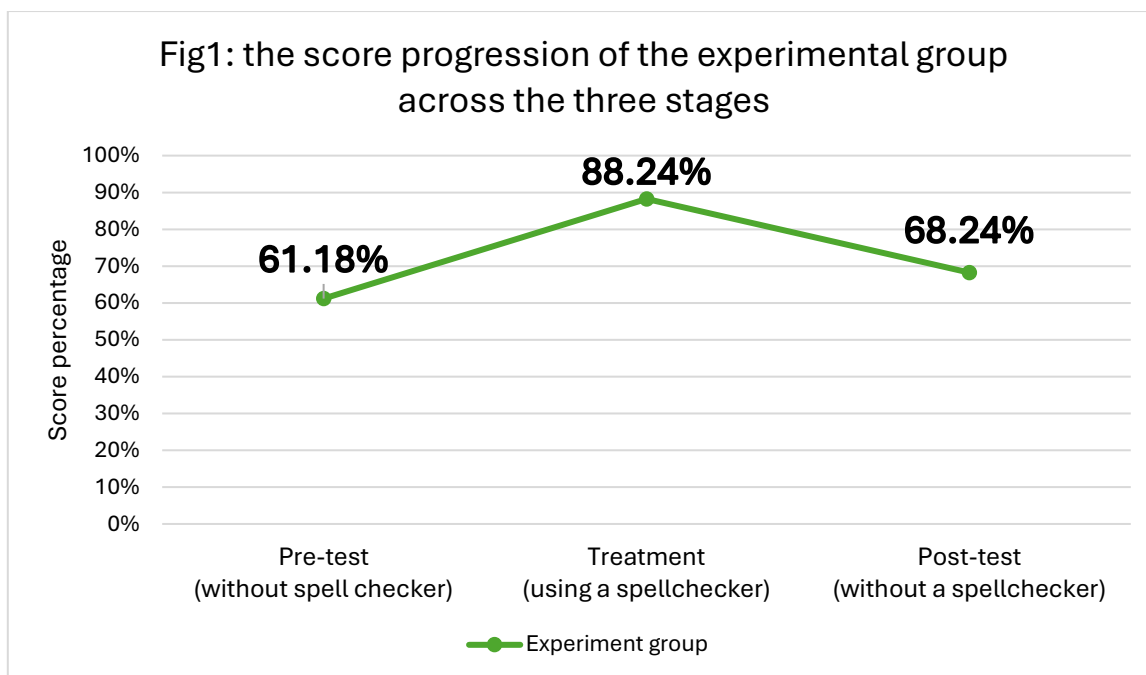


Figure 1: the score progression of the experimental group across the three stages

Fig1 is a line graph that illustrates the scoring Progression of the Experimental Group during the Pre-test, Treatment, and Post-test phases. The result analysis shows that the experiment group had a mean score of 61.18% in the Pre-test before using a spellchecker, this score has increased significantly to 88.24%, which emphasized the immediate impact of using a spell checker on writing accuracy. The post-test retained a mean score of 68.24% which is less than the treatment phase, yet higher than the pretest phase; although it decreased from the treatment phase, this score is still above the pre-test mean score.

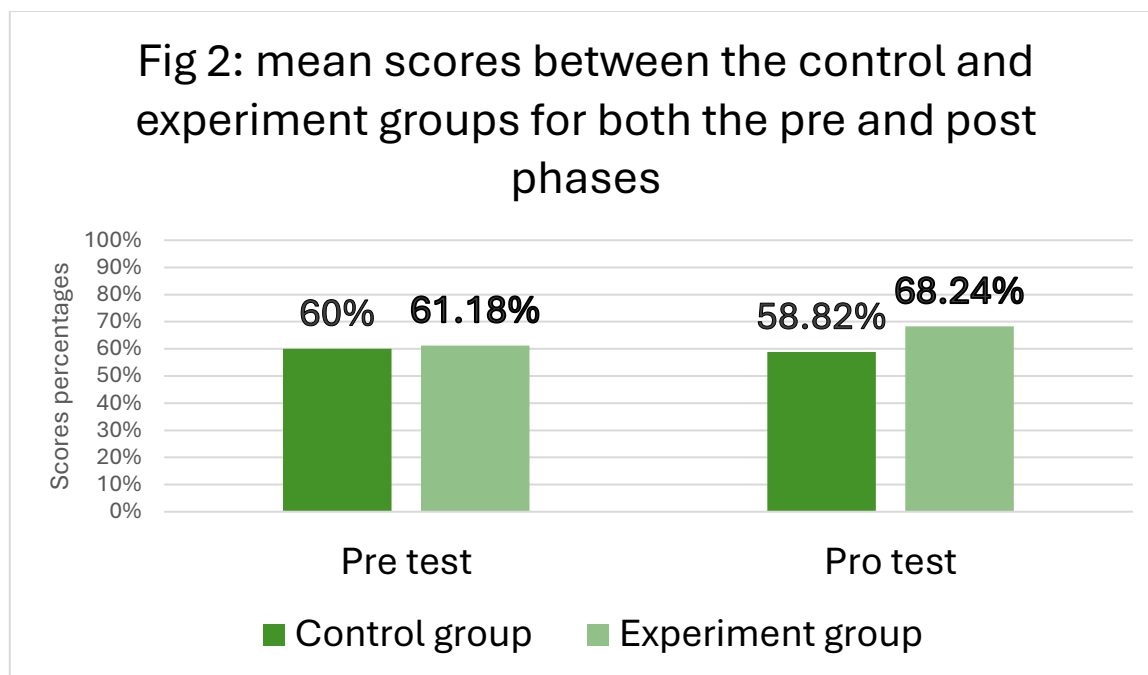


Figure 2: mean scores between the control and experiment groups for both the pre and post phases

Figure 2 is a bar chart that compares the mean scores between the control and experimental groups for both the pre-test and post-test phases. In the pre-test phase, the experimental group had a mean score of 61.18%, while the control group had a similar mean score of 60.00%, which indicates comparable spelling abilities between the two groups in the pre-test phase. In the post-test phase, the experimental group's mean score increased to 68.24% after using Grammarly during the treatment phase, while the control group's mean score decreased slightly to 58.82%. This comparison shows the positive impact of the spell checker on the experimental group's spelling correction abilities, while the control group, which did not receive any intervention, showed a stable lower performance.

Chapter 5: Discussion

After the results presented above, it is evident that using the spell checker has helped AUC students to spell correctly because the mean of the experiment group has a significant increase in the mean scores of the post-test compared to their performance in the pre-test. This result contradicts the results found in the research of Rimbar (2017), which found that students were not able to generate repairs of spelling errors in the post-test after using the spell checker. The result of the post-test shows also that although students were able to write better than the pre-test, most of the errors that occurred again were the common misspelling words, which were identified as the most misspelled words among students according to Oxford University Press (2012). On top of that, the most frequent misspelled word is “maneuver” with 37 out of 40 times excluding the treatment phase, spellings such as “manouver”, “manover”, “manevor” shows that AUC students struggle with phonetics and vowels placements. These findings match with the findings of Lin (2017) who argues that phonetics is usually challenging for students to spell because it requires convenience, memory, and effort to be spelled correctly. Surprisingly, a very interesting result shows that 6/7 of the students who spelled “questionnaire” in the pre-test correctly were from North Africa (Morocco, Algeria, and Tunisia) where they have a good background in the French language; because the word questionnaire is spelled the same in both languages, with an easy and more obvious spelling in the French language. These findings confirm the cross-linguistic influence and the theory that students always rely on their mother tongue/proficient languages to spell words correctly (Odlin, 1989; Schwartz, 1998). The notable rise in scores from the treatment phase to the post-test phase is another crucial finding that needs further discussion. The brief time difference between these two stages implies that spelling knowledge was most likely retained by students using short-term memory (STM). This behavior is consistent with the results of Margo et al. (2021), who found that short-term memory (STM) is generally more accurate at spelling than long-term memory (LTM). This study argues that STM may have helped students remember and replicate the spelling patterns and corrections they came across throughout the treatment phase.

Chapter 6: Conclusion

6.0 Summary:

This study examined the impact of spell checker usage on the spelling abilities of AUC students. The findings support the hypothesis that using spell checkers helps AUC students spell correctly, as demonstrated by the significant increase in mean scores from the pre-test to the post-test. However, certain common misspellings persisted, indicating areas where spell checkers might not be fully effective.

6.1 Theoretical Implications:

The findings support the theory of cross-linguistic influence, as demonstrated by the performance of North African students with a background in French. The study adds to the body of research on the effectiveness of spell checkers in language learning, particularly in spelling improvement.

6.2 Practical Implications:

Findings have shown poor spelling skills for AUC students, a point that has to be taken into consideration in order to develop English courses such as the critical writing course. Additional strategies may be necessary to address persistent spelling errors that are not easily corrected by spell checkers.

6.3 limitations and Future Research Recommendations:

The study was limited to AUC students who have taken the critical writing course, which may not be representative of the broader student population. The duration of the study was relatively short (less than a week), limiting insights into the long-term effectiveness of the spell checker. Furthermore, the study did not account for individual differences in learning styles and prior knowledge, which could influence the outcomes. The short time between the treatment and the post-test may have led to short-term memory effects, and the tests conducted individually in different areas may have introduced external factors affecting the results. Future research should investigate the long-term effects of spell checker use on spelling skills. Research should consider a more diverse sample to determine if the findings are generalizable. Additionally, future studies should allow more time between treatment and post-test to assess long-term learning effects. Conducting all tests in the same environment could help control for external factors that might influence the results.

6.4 Thesis summary:

In conclusion, the use of spell checkers significantly improved the spelling performance of AUC students, although certain common misspellings persisted. The study also indicates the importance of considering cross-linguistic influences and suggests that while technological tools are beneficial, they should be complemented with additional instructional strategies to address persistent challenges in spelling. Furthermore, addressing the identified gaps can improve the applicability of future research in this area.

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Appendix I:

The dictation test phrases:

-The **entrepreneur** had to **maneuver** through -a lot of **bureaucracy** to get his startup off the ground, which was **particularly challenging**.

-Despite the **unnecessary delays**, the **occurrence** of such obstacles didn't **embarrass** him; he was **prepared** to handle them.

-She was **referred** to a specialist, who asked her to fill out a detailed **questionnaire**, which she found **acceptable** despite the **inconvenience**.

-It is important to **separate** work and personal life to maintain balance, a point that is **definitely emphasized** by **successful** leaders.